

Approximate Counting of k -Paths in $O^*(2^k)$ Time

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Abstract

We resolve the conjecture of Koutis and Williams that k -paths can be approximately counted in $O^*(2^k)$ time. For a simple directed graph with n vertices and m arcs, our randomized algorithm returns a $(1 \pm \varepsilon)$ -approximation with failure probability at most δ in $O^*(2^k \varepsilon^{-2} \log(1/\delta))$ time.

1 Introduction

A directed k -path is an ordered tuple $(v_1, \dots, v_k)$ of distinct vertices such that (v_i, v_{i+1}) is an arc for every $1 \leq i < k$. The k -path problem asks whether a given graph contains such a path. It is a benchmark problem in parameterized algorithms [6, 15]: after the early $k! \text{poly}(n)$ -time algorithm of Monien and the color-coding method of Alon, Yuster, and Zwick [3, 17], algebraic fingerprints led to randomized algorithms with running times $O^*(2^{3k/2})$ [11] and $O^*(2^k)$ for directed graphs [21]. Here and throughout, O^* suppresses factors polynomial in the parameter and the input size, but not the displayed dependence on ε and δ . In contrast, exactly counting k -paths is $\#W[1]$ -hard for both directed and undirected graphs [10]. This gap raises a natural intermediate question.

Can directed k -paths be approximately counted in $O^*(2^k)$ time?

The existence of such an algorithm was conjectured by Koutis and Williams [12]. Color-coding gives a randomized $(1 \pm \varepsilon)$ -approximation in $O((2e)^k m \varepsilon^{-2})$ time for constant success probability [1]. Extensor-coding subsequently gave a faster algorithm with a running time of $O^*(4^k)$ [6]. Using representative weight functions, Lokshtanov, Saurabh, and Zehavi improved the running time to $O((2.619^k + n^{o(1)}) \varepsilon^{-2} (n + m))$ [15]. We resolve the conjecture of Koutis and Williams:

Theorem 1. *Let $k \geq 1$ and $0 < \varepsilon, \delta < 1$. Given a simple directed graph G , there is a randomized algorithm that returns a $(1 \pm \varepsilon)$ -approximation to the number of directed k -paths in G with success probability at least $1 - \delta$ and runs in $O^*(2^k \varepsilon^{-2} \log(1/\delta))$ time.*

The k -path algorithm is a consequence of a more general theorem for counting multilinear monomials in polynomials computed by homogeneous 0–1 right-skew circuits. Associate a directed walk $(v_1, \dots, v_k)$ with the noncommutative monomial $x_{v_1} \cdots x_{v_k}$. The walk is a path exactly when this monomial is multilinear, meaning that no variable occurs twice. Noncommutativity preserves the vertex order, so different walks do not collapse into the same monomial.

We call a noncommutative polynomial a 0–1 *polynomial* if all its coefficients belong to $\{0, 1\}$. For such a polynomial P , let $N(P)$ be the number of its multilinear monomials. A homogeneous 0–1 right-skew circuit is built from constants 0 and 1, addition, and product gates that multiply a previously computed polynomial from the right by one variable. Every gate is required to compute a 0–1 polynomial. See Section 2 for the formal model.

Theorem 2. *Let P be a noncommutative 0–1 polynomial of degree $k \geq 1$ computed by a homogeneous 0–1 right-skew circuit C . For $0 < \varepsilon, \delta < 1$, there is a randomized algorithm that, given C , returns $\hat{N}$ such that $\Pr[|\hat{N} - N(P)| \leq \varepsilon N(P)] \geq 1 - \delta$ and runs in $O^*(2^k \varepsilon^{-2} \log(1/\delta))$ time.*

Related work. Arvind and Raman gave early randomized parameterized counting approximation algorithms for patterns of bounded treewidth, including paths [4]. Alon and Gutner later used balanced families of perfect hash functions to derandomize color-coding and obtain deterministic multiplicative approximations for counting paths [2].

Using lifted labels in the exterior algebra on $\mathbb{Z}^{2k}$, Brand, Dell, and Husfeldt gave a randomized $O^*(4^k \varepsilon^{-2})$ -time approximation for directed k -path [6]. Lokshtanov, Björklund, Saurabh, and Zehavi subsequently obtained polynomial-space algorithms for directed and undirected k -path counting. Their randomized running time was $O^*(4^{k+O(\log k(\log k + \log(1/\varepsilon)))})$, whereas their deterministic running time was $O^*(4^{k+O(\sqrt{k}(\log^2 k + \log^2(1/\varepsilon))))$ [14].

Lokshtanov, Saurabh, and Zehavi [15] used representative weight functions to obtain the $O^*(2.619^k \varepsilon^{-2})$ -time approximation for k -path. More generally, for a homogeneous degree- k commutative polynomial represented by a monotone arithmetic circuit C , they gave a randomized $(1 \pm \varepsilon)$ -approximation to the sum of the coefficients of its multilinear monomials in $O^*(3.841^k \varepsilon^{-6})$ time. Pratt gave a randomized constant-success $(1 \pm \varepsilon)$ -approximation for the corresponding coefficient sum of a homogeneous commutative polynomial with nonnegative real coefficients, given by black-box evaluation, in $O^*(4.075^k \varepsilon^{-2} \log(1/\varepsilon))$ time and polynomial space [20].

For the decision problem, group-algebra fingerprints annihilated nonmultilinear terms and yielded randomized multilinear detection algorithms [11, 12, 21]. Brand and Pratt developed a symbolic-differentiation framework for skew circuits based on determinants [7]. Determinantal sieving detected multilinear monomials whose supports formed bases of a linear matroid. Over general fields, it evaluated a circuit over an exterior algebra [9]. Dynamic representative sets gave a deterministic $O^*(2^{k+O(\sqrt{k} \log^2 k)})$ -time algorithm for weighted directed k -path in the word-RAM model, with every edge weight stored in one word [18]. For undirected k -path, narrow sieves gave a polynomial-space randomized $O^*(1.66^k)$ -time decision algorithm with constant one-sided error [5].

Organization. Section 2 formalizes the polynomial, tensor, circuit, and exterior-algebra notation used later. Section 3.1 defines the estimator and proves its approximation guarantee. Section 3.2 gives its circuit implementation and proves Theorem 2. Section 4 derives Theorems 1 and 5.

2 Preliminaries

Polynomials and tensors. Fix $n \geq 1$. For a positive integer r , let $[r] = \{1, \dots, r\}$. We work in the noncommutative polynomial ring $\mathbb{Z}\langle x_1, \dots, x_n \rangle$. For every positive integer i , each degree- i monomial is uniquely indexed by a tuple $\tau = (\tau_1, \dots, \tau_i) \in [n]^i$, and we write $x_\tau = x_{\tau_1} \cdots x_{\tau_i}$. The monomial x_τ is *multilinear* if the entries of τ are pairwise distinct.

A homogeneous degree- i polynomial has the form $P = \sum_{\tau \in [n]^i} c_\tau x_\tau$. It is a 0–1 polynomial if $c_\tau \in \{0, 1\}$ for every $\tau \in [n]^i$. For such a polynomial, define $N(P) = \sum_{\tau \in [n]^i: x_\tau \text{ multilinear}} c_\tau$. The polynomial P can equivalently be represented by its coefficient tensor $(c_\tau)_{\tau \in [n]^i} \in \mathbb{Z}^{[n]^i}$. More generally, a real tensor $A \in \mathbb{R}^{I_1 \times \cdots \times I_r}$ is an array indexed by finite sets $I_1, \dots, I_r$. Its Frobenius norm is

$$\|A\|_F = \left(\sum_{(a_1, \dots, a_r) \in I_1 \times \cdots \times I_r} A(a_1, \dots, a_r)^2 \right)^{1/2}.$$

To *flatten* a tensor with specified row coordinates means to view the same entries as a matrix whose row index is the Cartesian product of these coordinates and whose column index is the Cartesian product of the remaining coordinates. Flattening only reindexes the entries and therefore preserves the Frobenius norm.

Circuits. A *homogeneous 0–1 right-skew circuit* is a directed acyclic graph with a designated output gate. Each gate is a constant input gate labeled 0 or 1, a binary addition gate $g = h_1 + h_2$, or a product gate $g = hx_j$ with $j \in [n]$, where h, h_1, h_2 are predecessor gates of g . These operations recursively define the noncommutative polynomial computed at each gate, and we require every such polynomial to be a 0–1 polynomial. Every gate also has a syntactic degree. Constants have degree zero, the two inputs and the output of an addition gate have the same degree, and hx_j has degree one more than h .

Exterior algebra. We adapt the concrete presentation of exterior algebra from Brand, Dell, and Husfeldt [6, Section 2] to the coefficient ring $\mathbb{Z}$. Let $e_1, \dots, e_k$ be the standard basis of $\mathbb{Z}^k$. For every $I \subseteq [k]$, let e_I be a formal basis element, where $e_\emptyset = 1$ and $e_{\{j\}} = e_j$. For $0 \leq r \leq k$, the r th *exterior power* $\Lambda^r(\mathbb{Z}^k)$ is the free $\mathbb{Z}$ -module with basis $\{e_I : I \subseteq [k], |I| = r\}$. The *exterior algebra* is $\Lambda(\mathbb{Z}^k) = \bigoplus_{r=0}^k \Lambda^r(\mathbb{Z}^k)$. We identify $\mathbb{Z}^k$ with $\Lambda^1(\mathbb{Z}^k)$ through the basis $e_1, \dots, e_k$.

The *wedge product* is the $\mathbb{Z}$ -bilinear multiplication determined on basis elements by

$$e_I \wedge e_J = \begin{cases} 0, & I \cap J \neq \emptyset, \\ (-1)^{|\{(a,b) \in I \times J : a > b\}|} e_{I \cup J}, & I \cap J = \emptyset. \end{cases}$$

It is associative, respects the grading $\Lambda^r(\mathbb{Z}^k) \wedge \Lambda^s(\mathbb{Z}^k) \subseteq \Lambda^{r+s}(\mathbb{Z}^k)$, and is alternating on vectors. In particular, $y \wedge y = 0$ for every $y \in \mathbb{Z}^k$. If $I = \{i_1 < \dots < i_r\}$, then $e_I = e_{i_1} \wedge \dots \wedge e_{i_r}$. Thus $\Lambda^r(\mathbb{Z}^k)$ has $\binom{k}{r}$ coordinates and the full exterior algebra has 2^k coordinates. Finally, for $y_1, \dots, y_k \in \mathbb{Z}^k$, expanding the wedge product gives

$$y_1 \wedge \dots \wedge y_k = \det[y_1 \ \dots \ y_k] e_{[k]}.$$

3 The algorithm

In this section, we prove [Theorem 2](#).

Theorem 2. *Let P be a noncommutative 0–1 polynomial of degree $k \geq 1$ computed by a homogeneous 0–1 right-skew circuit C . For $0 < \varepsilon, \delta < 1$, there is a randomized algorithm that, given C , returns $\hat{N}$ such that $\Pr[|\hat{N} - N(P)| \leq \varepsilon N(P)] \geq 1 - \delta$ and runs in $O^*(2^k \varepsilon^{-2} \log(1/\delta))$ time.*

Throughout this section, fix a polynomial P and a homogeneous 0–1 right-skew circuit C computing it as in [Theorem 2](#), with variables $x_1, \dots, x_n$. For every $\tau \in [n]^k$, let c_τ be the coefficient of x_τ in P . Let $N = N(P)$, set $\ell = 2k$, and fix $0 < \varepsilon, \delta < 1$. In this section, O^* suppresses factors polynomial in k and in the encoding size of C , but not the displayed dependence on ε and δ .

Our algorithm returns 0 whenever $N = 0$, so we assume $N > 0$ below.

We call a random variable Y an *efficient estimator* of N if $\mathbb{E}(Y) = N > 0$ and $\mathbb{E}(Y^2)/(\mathbb{E}(Y))^2$ is bounded by $k^{O(1)}$. The standard median-of-means bound shows that

$$O\left(\frac{\mathbb{E}(Y^2)}{(\mathbb{E}(Y))^2} \varepsilon^{-2} \log(1/\delta)\right)$$

independent samples of Y suffice to compute $\hat{N}$ such that $\Pr[|\hat{N} - N| > \varepsilon N] \leq \delta$. See, for example, Mitzenmacher and Upfal [16, Sections 3.3–3.4]. It therefore suffices to construct an efficient estimator of N and compute each sample in $O^*(2^k)$ time.

3.1 Constructing an efficient estimator

Let $\sigma_{j,r}$, $j \in [n]$ and $r \in [k]$, be mutually independent random variables, each uniform on $\{-1, 1\}$, and define $\sigma_j = (\sigma_{j,1}, \dots, \sigma_{j,k})^\top \in \mathbb{Z}^k$. For every $\tau = (\tau_1, \dots, \tau_k) \in [n]^k$, let $D_\tau = \det[\sigma_{\tau_1} \cdots \sigma_{\tau_k}]$. Define $\Psi \in \mathbb{R}^{[n]^k}$ by $\Psi(\tau) = c_\tau D_\tau$. Recall that its Frobenius norm is $\|\Psi\|_F = (\sum_{\tau \in [n]^k} \Psi(\tau)^2)^{1/2}$. Since $c_\tau \in \{0, 1\}$ for every $\tau \in [n]^k$, we have

$$\|\Psi\|_F^2 = \sum_{\tau \in [n]^k} c_\tau^2 D_\tau^2 = \sum_{\tau \in [n]^k} c_\tau D_\tau^2.$$

If x_τ is not multilinear, then the matrix defining D_τ has two equal columns, so $D_\tau = 0$.

We first show that the normalized squared Frobenius norm $\|\Psi\|_F^2/k!$ is an efficient estimator of N . The algorithm of Brand, Dell, and Husfeldt uses this estimator [6, Section 3.6]. We include the proof in our setting for completeness.

Lemma 3. $\mathbb{E}(\|\Psi\|_F^2) = k!N$ and $\mathbb{E}(\|\Psi\|_F^4) \leq (k!)^2 k^3 N^2$.

Proof. Fix $\tau \in [n]^k$ such that x_τ is multilinear. Since $\tau_1, \dots, \tau_k$ are distinct, D_τ has the same distribution as $\det(B)$, where B is a $k \times k$ random sign matrix with independent entries uniform in $\{-1, 1\}$. The determinant-moment formulas of Nyquist, Rice, and Riordan [19] imply

$$\mathbb{E}(\det(B)^2) = k! \quad \text{and} \quad \mathbb{E}(\det(B)^4) \leq (k!)^2 k^3.$$

Let $\mathcal{M}$ be the set of $\tau \in [n]^k$ such that $c_\tau = 1$ and x_τ is multilinear. Then $|\mathcal{M}| = N$. Since $D_\tau = 0$ when x_τ is not multilinear,

$$\|\Psi\|_F^2 = \sum_{\tau \in \mathcal{M}} D_\tau^2.$$

Taking expectations gives $\mathbb{E}(\|\Psi\|_F^2) = k!N$. Moreover, Cauchy-Schwarz gives

$$\|\Psi\|_F^4 = \left(\sum_{\tau \in \mathcal{M}} D_\tau^2 \right)^2 \leq \left(\sum_{\tau \in \mathcal{M}} 1 \right) \left(\sum_{\tau \in \mathcal{M}} D_\tau^4 \right) = N \sum_{\tau \in \mathcal{M}} D_\tau^4.$$

Taking expectations and using $\mathbb{E}(D_\tau^4) \leq (k!)^2 k^3$ for every $\tau \in \mathcal{M}$ gives $\mathbb{E}(\|\Psi\|_F^4) \leq (k!)^2 k^3 N^2$. $\square$

The extensor-coding construction of Brand, Dell, and Husfeldt computes $\|\Psi\|_F^2$ in $O^*(4^k)$ arithmetic operations [6]. It uses *lifts* that map each $\sigma_j \in \mathbb{Z}^k \subseteq \Lambda(\mathbb{Z}^k)$ to an element of $\Lambda^2(\mathbb{Z}^{2k})$ and evaluates C over $\Lambda(\mathbb{Z}^{2k})$. We instead compress Ψ using random sign matrices.

Compression by random sign matrices. Independently of the vectors $\sigma_1, \dots, \sigma_n$, let $R_1 \in \{-1, 1\}^{[\ell] \times [n]}$ and $R_i \in \{-1, 1\}^{[\ell] \times ([\ell] \times [n])}$ for every $2 \leq i \leq k$ be random matrices whose entries are mutually independent and uniform. Flatten Ψ with its first coordinate as the row coordinate, multiply it from the left by R_1 , and let the resulting tensor be $S_1 \in \mathbb{R}^{[\ell] \times [n]^{k-1}}$. Here and below, a factor $[n]^0$ is omitted. For each $i = 2, \dots, k$, flatten $S_{i-1} \in \mathbb{R}^{[\ell] \times [n]^{k-i+1}}$ with its first two coordinates as the row coordinates, multiply it from the left by R_i , and denote the result by $S_i \in \mathbb{R}^{[\ell] \times [n]^{k-i}}$. Thus $S_k \in \mathbb{R}^{[\ell]} \cong \mathbb{R}^\ell$. For every finite-dimensional real vector y , write $\|y\|_2^2$ for the sum of the squares of its coordinates. The final vector is given, for every $a_k \in [\ell]$, by

$$S_k(a_k) = \sum_{\tau \in [n]^k} \Psi(\tau) \sum_{a_1, \dots, a_{k-1} \in [\ell]} R_1[a_1, \tau_1] \left(\prod_{t=2}^k R_t[a_t, (a_{t-1}, \tau_t)] \right). \quad (3.1)$$

We now verify that S_k yields an efficient estimator of N . The key point is that, after normalization, successive compression by random sign matrices preserves the expected squared Frobenius norm and increases its fourth moment by only a constant factor.

Lemma 4. $\|S_k\|_2^2/(\ell^k k!)$ is an efficient estimator of N .

Proof. We use the following one-step estimate.

Claim. Let d and t be positive integers. Let X be a random matrix in $\mathbb{R}^{d \times t}$, and let $R \in \{-1, 1\}^{\ell \times d}$ be a random sign matrix independent of X . Then

$$\mathbb{E}(\|RX\|_F^2) = \ell \mathbb{E}(\|X\|_F^2) \quad \text{and} \quad \mathbb{E}(\|RX\|_F^4) \leq \ell^2(1 + 2/\ell)\mathbb{E}(\|X\|_F^4).$$

Proof. First suppose that $X \in \mathbb{R}^{d \times t}$ is fixed.

Define $M = XX^\top$, and write $\text{tr } M$ for the sum of its diagonal entries. Let $r_1, \dots, r_\ell \in \{-1, 1\}^d$ be the rows of R , and define $Q_j = r_j^\top M r_j$ for every $j \in [\ell]$. The j th row of RX is $r_j^\top X$. Therefore,

$$\|RX\|_F^2 = \sum_{j=1}^{\ell} \|r_j^\top X\|_2^2 = \sum_{j=1}^{\ell} r_j^\top X X^\top r_j = \sum_{j=1}^{\ell} Q_j.$$

Since M is symmetric and the coordinates of r_j have mean zero and square one,

$$Q_j = \text{tr } M + 2 \sum_{1 \leq u < v \leq d} M_{uv} r_{j,u} r_{j,v} \quad \text{and} \quad \mathbb{E}(Q_j) = \text{tr } M.$$

When the centered expression is squared, every cross term indexed by two distinct pairs has expectation zero because some coordinate occurs exactly once. Hence

$$\text{Var}(Q_j) = \mathbb{E}((Q_j - \mathbb{E}(Q_j))^2) = 4\mathbb{E}\left(\left(\sum_{1 \leq u < v \leq d} M_{uv} r_{j,u} r_{j,v}\right)^2\right) = 4 \sum_{1 \leq u < v \leq d} M_{uv}^2 \leq 2\|M\|_F^2.$$

Let $\lambda_1, \dots, \lambda_d \geq 0$ be the eigenvalues of the positive semidefinite matrix M . Then

$$\|M\|_F^2 = \text{tr}(M^2) = \sum_{h=1}^d \lambda_h^2 \leq \left(\sum_{h=1}^d \lambda_h\right)^2 = (\text{tr } M)^2,$$

and $\text{tr } M = \text{tr}(XX^\top) = \|X\|_F^2$. Since the Q_j are independent copies, linearity of expectation and additivity of variance give $\mathbb{E}(\|RX\|_F^2) = \ell \text{tr } M = \ell \|X\|_F^2$ and $\text{Var}(\|RX\|_F^2) \leq 2\ell(\text{tr } M)^2$. Therefore,

$$\mathbb{E}(\|RX\|_F^4) = \text{Var}(\|RX\|_F^2) + (\mathbb{E}(\|RX\|_F^2))^2 \leq 2\ell \|X\|_F^4 + \ell^2 \|X\|_F^4 = \ell^2(1 + 2/\ell)\|X\|_F^4.$$

Since X is a random matrix independent of R , conditioning on X gives $\mathbb{E}(\|RX\|_F^2) = \ell \mathbb{E}(\|X\|_F^2)$. It also gives $\mathbb{E}(\|RX\|_F^4) \leq \ell^2(1 + 2/\ell) \mathbb{E}(\|X\|_F^4)$. $\square$

We apply the claim at each flattening. Let $S_0 = \Psi$. For each $i \in [k]$, the matrix multiplied by R_i is a flattening of S_{i-1} . It depends only on $\sigma_1, \dots, \sigma_n, R_1, \dots, R_{i-1}$ and is therefore independent of R_i . Since flattening preserves the Frobenius norm, applying the preceding bounds inductively gives, for every $i \in [k]$,

$$\begin{aligned} \mathbb{E}(\|S_i\|_F^2) &= \ell \mathbb{E}(\|S_{i-1}\|_F^2), \\ \mathbb{E}(\|S_i\|_F^4) &\leq \ell^2(1 + 2/\ell) \mathbb{E}(\|S_{i-1}\|_F^4). \end{aligned}$$

By iterating from $S_0 = \Psi$, using $\|S_k\|_2 = \|S_k\|_F$, and applying [Lemma 3](#), we obtain

$$\begin{aligned} \mathbb{E}(\|S_k\|_2^2/(\ell^k k!)) &= \mathbb{E}(\|\Psi\|_F^2)/k! = N, \\ \mathbb{E}((\|S_k\|_2^2/(\ell^k k!))^2) &\leq \left(1 + \frac{2}{\ell}\right)^k k^3 N^2 = \left(1 + \frac{1}{k}\right)^k k^3 N^2. \end{aligned}$$

Since $(1 + 1/k)^k \leq e$, the lemma follows. $\square$

By [Lemma 4](#) and the median-of-means bound above, $O^*(\varepsilon^{-2} \log(1/\delta))$ independent trials suffice to return $\hat{N}$ such that $\Pr[|\hat{N} - N| > \varepsilon N] \leq \delta$. Thus it remains only to compute each sample within the claimed running time.

3.2 Sampling from the efficient estimator

Use the random vectors $\sigma_1, \dots, \sigma_n$ and random matrices $R_1, \dots, R_k$ defined in [Section 3.1](#). Evaluate the ancestors of the output gate over the exterior algebra $\Lambda(\mathbb{Z}^k)$ from [Section 2](#).

For every gate g of degree $i \geq 1$ and every $\tau \in [n]^i$, let $c_g(\tau)$ be the coefficient of x_τ in the noncommutative polynomial computed at g ; since C is a 0–1 circuit, $c_g(\tau) \in \{0, 1\}$. The values below store signed sums of these monomial contributions after replacing variables by their exterior-algebra labels. Evaluate these gates in topological order. Degree-zero gates store their values in $\{0, 1\}$. For every gate g of degree $i \geq 1$ and every $a \in [\ell]$, store $F_{g,a} \in \Lambda^i(\mathbb{Z}^k)$. Addition gates are evaluated componentwise. If $g = hx_j$ has degree one, we may assume that h stores 1, since otherwise g computes zero and can be ignored. Define $F_{g,a} = R_1[a, j]\sigma_j$. If $g = hx_j$ has degree $i \geq 2$, define $F_{g,a} = (\sum_{b=1}^\ell R_i[a, (b, j)]F_{h,b}) \wedge \sigma_j$.

For every gate g of degree $i \geq 1$ and every $a_i \in [\ell]$, we claim that

$$F_{g,a_i} = \sum_{\tau \in [n]^i} c_g(\tau) \sum_{a_1, \dots, a_{i-1} \in [\ell]} R_1[a_1, \tau_1] \left(\prod_{t=2}^i R_t[a_t, (a_{t-1}, \tau_t)] \right) \cdot \sigma_{\tau_1} \wedge \dots \wedge \sigma_{\tau_i}. \quad (3.2)$$

For $i = 1$, the inner sum is omitted and the product is interpreted as 1.

We prove (3.2) by induction over the gates. At a degree-one product gate $g = hx_j$, we may assume that h stores 1. The coefficient of x_j is then one, all other coefficients are zero, and $F_{g,a_1} = R_1[a_1, j]\sigma_j$. At an addition gate $g = h_1 + h_2$, both the coefficients and the stored elements are the sums of the corresponding values at h_1 and h_2 .

Now let $g = hx_j$ have degree $i \geq 2$. A monomial $x_\tau = x_{\tau_1} \dots x_{\tau_i}$ has nonzero coefficient at g only if $\tau_i = j$, and in this case $c_g(\tau) = c_h((\tau_1, \dots, \tau_{i-1}))$. Substituting the expansion of each $F_{h,b}$ into the recurrence makes the sum over b the sum over a_{i-1} , introduces the factor $R_i[a_i, (a_{i-1}, \tau_i)]$, and appends σ_{τ_i} to the wedge product. This proves (3.2).

Let o be the output gate. Since $\Psi(\tau) = c_\tau D_\tau$, $c_o(\tau) = c_\tau$, and $\sigma_{\tau_1} \wedge \dots \wedge \sigma_{\tau_k} = D_\tau e_{[k]}$, (3.1) and (3.2) give, for every $a_k \in [\ell]$,

$$F_{o,a_k} = \sum_{\tau \in [n]^k} c_\tau D_\tau \sum_{a_1, \dots, a_{k-1} \in [\ell]} R_1[a_1, \tau_1] \left(\prod_{t=2}^k R_t[a_t, (a_{t-1}, \tau_t)] \right) e_{[k]} = S_k(a_k) e_{[k]}.$$

Thus the stored output coefficients determine $\|S_k\|_2^2 / (\ell^k k!) = (\ell^k k!)^{-1} \sum_{a=1}^\ell S_k(a)^2$ using only rational arithmetic.

Running time. Let L be the encoding size of C . After relabeling the variables occurring in C , we may assume that $n \leq L$. By [Section 2](#), an element of $\Lambda^i(\mathbb{Z}^k)$ has at most 2^k coefficients. Since $\ell = 2k$, every gate can be evaluated using $2^k \text{poly}(k)$ integer operations. For a degree- i gate, each coefficient in (3.2) is a sum of at most $L^i \ell^{i-1}$ terms, each of which is a coefficient of an i -fold wedge of sign vectors and hence has absolute value at most $i!$. Thus every integer computed in one trial has $O(k(1 + \log L + \log k))$ bits. Squaring and summing the output coefficients changes this bound only by a constant factor. Therefore every integer operation within one trial takes time polynomial in k and L , and one trial takes $O^*(2^k)$ time. Each estimator value has denominator $\ell^k k!$, and averaging $O(k^3 \varepsilon^{-2})$ values adds only $O(1 + \log k + \log(1/\varepsilon))$ bits to its numerator and denominator. The group averages have a common denominator, so their median can be selected using a linear number of comparisons between their numerators. These comparisons and the remaining amplification arithmetic take polynomial time. The amplification in [Section 3.1](#) uses $O^*(\varepsilon^{-2} \log(1/\delta))$ trials, giving a total running time of $O^*(2^k \varepsilon^{-2} \log(1/\delta))$.

4 Applications

For completeness, we give the noncommutative-polynomial construction underlying [Theorem 1](#). We also derive, by the same approach, a result for set packing.

4.1 Directed paths

A directed k -path is an ordered tuple $(v_1, \dots, v_k)$ of distinct vertices such that (v_i, v_{i+1}) is an arc for every $1 \leq i < k$.

Theorem 1. *Let $k \geq 1$ and $0 < \varepsilon, \delta < 1$. Given a simple directed graph G , there is a randomized algorithm that returns a $(1 \pm \varepsilon)$ -approximation to the number of directed k -paths in G with success probability at least $1 - \delta$ and runs in $O^*(2^k \varepsilon^{-2} \log(1/\delta))$ time.*

Proof. Let the input digraph have vertex set $[n]$ and arc set A , where $|A| = m$. If $n < k$, return 0, so assume $n \geq k$. Define $Z_0 = 0$ and $Z_i = Z_{i-1}x_1$ for $i \in [k]$; thus Z_i is a syntactically homogeneous zero polynomial of degree i . For every $v \in [n]$, define $P_{v,1} = x_v$. For $i = 2, \dots, k$ and $v \in [n]$, define

$$P_{v,i} = \left(\sum_{u:(u,v) \in A} P_{u,i-1} \right) x_v,$$

where an empty sum is represented by Z_{i-1} . The polynomial $P_{v,i}$ stores the directed walks on i vertices that end at v .

We prove this claim by induction on i . It holds for $i = 1$. For $i \geq 2$, each summand indexed by an arc (u, v) appends v to a walk ending at u . Conversely, deleting the final vertex of a walk ending at v identifies its summand. The monomial records the entire vertex sequence and therefore occurs with coefficient one.

Let $P_k = \sum_{v \in [n]} P_{v,k}$. The last variable identifies the endpoint, so every monomial of P_k still has coefficient one. Its monomials encode all directed walks on k vertices, and a monomial is multilinear exactly when its variables are distinct. Thus the multilinear monomials of P_k are in bijection with the directed k -paths.

Every intermediate polynomial is 0–1: the predecessor sums combine disjoint supports, right multiplication is injective on monomials, and the final sum combines monomials with distinct last variables. Thus these recurrences and the final sum form a polynomial-size homogeneous 0–1 right-skew circuit. Applying [Theorem 2](#) to P_k gives the claimed approximation and running time. $\square$

4.2 Set packing

A k -packing is a k -element subfamily of pairwise disjoint sets. Counting k -matchings is $\#W[1]$ -hard even in bipartite graphs [8]. Liu, Wang, and Wang gave fixed-parameter randomized approximation schemes for counting packings in families of sets of fixed size [13]. For families of d -element sets, the representative-weight method gives a randomized $(1 \pm \varepsilon)$ -approximation in $O^*(2.619^{dk} \varepsilon^{-2})$ time [15]. The following theorem improves this exponential factor to 2^{dk} .

Theorem 5. *Let $d, k \geq 1$ and $0 < \varepsilon, \delta < 1$. Given a family $\mathcal{E}$ of $m \geq k$ distinct d -element sets, there is a randomized algorithm that returns a $(1 \pm \varepsilon)$ -approximation to the number of k -packings in $\mathcal{E}$ with success probability at least $1 - \delta$ and runs in $O^*(2^{dk} \varepsilon^{-2} \log(1/\delta))$ time.*

Proof. Let $\mathcal{E} = \{E_1, \dots, E_m\}$ be the input family. Index a variable x_u by every ground element u . For each $h \in [m]$, choose an ordering $u_{h,1}, \dots, u_{h,d}$ of E_h and define $M_h = x_{u_{h,1}} \cdots x_{u_{h,d}}$. We list selected sets in increasing input order because a commutative product would merge distinct subfamilies with the same union.

For every $h \in [m]$ and $0 \leq j \leq \min\{h, k\}$, the polynomial $P_{h,j}$ stores the j -element subfamilies of $\{E_1, \dots, E_h\}$. Define $P_{0,0} = 1$ and $P_{h,0} = 1$ for $h \in [m]$. For every $h \in [m]$ and $1 \leq j \leq \min\{h, k\}$, define

$$P_{h,j} = \begin{cases} P_{h-1,j} + P_{h-1,j-1}M_h, & j < h, \\ P_{h-1,h-1}M_h, & j = h. \end{cases}$$

Expanding each multiplication by M_h into d consecutive right-product gates implements these recurrences by a polynomial-size homogeneous right-skew circuit with output $P_{m,k}$.

Induction on h , using $P_{h,0} = 1$ as the base state, shows that for every $1 \leq j \leq \min\{h, k\}$, the monomials of $P_{h,j}$ are exactly $M_{i_1} \cdots M_{i_j}$ indexed by the sequences $1 \leq i_1 < \cdots < i_j \leq h$. The two terms of the recurrence correspond to subfamilies that omit or contain E_h , respectively. These cases are disjoint and exhaustive. Each such monomial has a unique decomposition into blocks of length d , and the input sets are distinct, so different subfamilies give different monomials. Each monomial has coefficient one. Each intermediate right-product gate appends a fixed prefix of M_h and therefore also computes a 0–1 polynomial. Hence the circuit above is a homogeneous 0–1 right-skew circuit.

Such a monomial is multilinear exactly when the selected sets are pairwise disjoint. Thus $P_{m,k}$ is a homogeneous 0–1 polynomial of degree dk , and $N(P_{m,k})$ is the number of k -packings in $\mathcal{E}$. Applying [Theorem 2](#) gives the claimed approximation and running time. $\square$

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Declaration of AI use

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